

MATHEMATICAL MODELING COURSE (CO2011)
SEMESTER 252 ASSIGNMENT

Invigilator Assignment Problem (IAP)

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Dear students,

For this semester's assignment in the Mathematical Modeling Course, your team (only allow 4-5) will apply Integer Linear Programming (ILP) to solve a real-world scheduling problem. This 2-week assignment requires you to mathematically model and solve the Invigilator Assignment Problem (IAP) using actual (anonymized) data from the Faculty of Computer Science and Engineering, HCMUT.

1. Problem Introduction & Context

In a university environment, scheduling exams and assigning invigilators (proctors) is one of the most complex and resource-intensive tasks each semester. The Invigilator Assignment Problem belongs to the class of Constraint Satisfaction Problems and is **NP-Hard** in nature.

Practical Significance & Complexity: In reality, schedulers face a massive volume of data: thousands of exam shifts spanning multiple weeks, hundreds of invigilators with overlapping schedules, and strict regulations. Manual scheduling often results in conflicts or highly unfair workload distributions.

A Toy Example: Consider a micro-scale scenario:

- **Set of Invigilators (I):** 3 staff members (CB1, CB2, CB3).
- **Set of Shifts (J):** 2 shifts (Morning, Afternoon). Each shift requires 2 invigilators.
- **Constraint:** CB1 is busy in the morning ($x_{11} = 0$).
- **Variables:** Let $x_{ij} \in \{0, 1\}$ (1 if staff i is assigned to shift j , 0 otherwise).
- **Objective:** Minimize variance in workload among staff.

References & Dataset:

- **Literature:** The primary reference is [Bakhtiar et al., 2015](#). You are encouraged to explore and cite other relevant academic literature.
- **Experimental Dataset & Baseline:** Use the anonymized dataset here: [Anonymized Dataset](#). **Important:** This file contains an existing schedule. You must treat this as your **BASELINE** solution. Extract parameters from this data and run your solver to improve upon it.

- **Simulated Location Preferences:** Categorize staff into 3 groups (with your own proposed settings):

1. Residing near Campus 1 (CS1) - prefer proctoring at CS1.
2. Residing near Campus 2 (CS2) - prefer proctoring at CS2.
3. Equidistant/Balanced location - no preference.

Incorporate these preferences into your model (e.g., as soft constraints or distance penalties).

2. The Ten (10) Assessment Requirements

1. **Parameters Analysis:** Extract and define all Sets and Parameters. Include simulated location data.
2. **Decision Variables:** Precisely define types (Binary, Integer, Continuous) and physical meaning.
3. **Objective Function:** Establish a function to maximize fairness or minimize travel penalties. Alternatively, teams may propose their own unique objectives, provided that they include clear and logical justifications for their practical relevance.
4. **Hard Constraints:** Formulate equations for inviolable rules (e.g., capacity, no double-booking).
5. **Soft Constraints:** Formulate equations for desirable rules (e.g., location preferences, fatigue limits).
6. **Data Preprocessing:** Write a Python script to read the dataset and generate synthetic preferences.
7. **Implementation of a solver:** Integrate the model into a free solver (PuLP, OR-Tools, or SCIP, etc) at your own programming language and tool preferences.
8. **Tuning:** Solve the model. If *Infeasible*, apply Relaxation or weight tuning.
9. **Performance Analysis:** Evaluate complexity, execution time, and compare against the baseline.
10. **Future Extension:** Propose alternative approaches (e.g., Genetic Algorithms) for large-scale scaling.

3. Submission Guidelines & Regulations

- **Platform:** ONLY the team leader submits a single .ZIP file (Report + Slides + Presentation Video + Code) via [HCMUT's LMS](#).
- **Language:** English-taught program students **MUST** write in English. Others may use Vietnamese or English.
- **Academic Integrity:** Violations (plagiarism, code copying) result in a **Grade of 0**.
- **Teamwork:** The appendix must include Meeting Minutes and a signed Contribution Table (%).

4. Additional Instructions & Rubric

Packaging: Folder must include `requirements.txt` and a `README.md`, together with a link to the video clip about 30 minutes of your team presentation (all members must present with slides and speaker's face appearance).

Assessment Criteria	Weight
Mathematical Modeling (Req. 1-5)	40%
Code & Solver Implementation (Req. 6-7)	30%
Analysis, Tuning & Extension (Req. 8-10)	20%
Report Quality & Presentation	10%

Good luck with your assignment. For any issues, post on the LMS course forum.